

Heart Disease Prediction

GitHub Repository Documentation (README.md)

This repository contains an end-to-end machine learning pipeline to predict the presence of heart disease based on clinical features. It includes data preprocessing, exploratory data analysis, multi-model evaluation, and hyperparameter tuning to deploy the highest-performing model.

Project Structure

```
Heart-Disease-Prediction/
├── data/                # Raw and processed datasets
├── notebooks/           # Jupyter notebooks for experimentation
├── models/              # Saved model (.pkl) and scaler files
├── src/                 # Source code scripts
├── app.py               # Application entry point
├── requirements.txt      # Project dependencies
└── README.md            # Project documentation
```

Dataset

The project uses the Cleveland Heart Disease dataset (`heart_cleveland_upload.csv`). The dataset contains clinical features like age, sex, chest pain type (cp), resting blood pressure, and cholesterol levels to predict the binary target condition (0 = healthy, 1 = heart disease).

Features & Implementation

1. Data Cleaning & Inspection

- Validates dataset shapes, column formats, and checks for missing or null values.
- Detects and drops duplicate rows automatically to maintain data integrity.

2. Exploratory Data Analysis (EDA)

- Visualizes target distribution to verify class balance.
- Maps feature distributions using frequency histograms.
- Generates a correlation matrix heat map to identify dependencies between variables.

3. Preprocessing

- Splits data into an 80/20 train-test ratio, stratified by the target label.
- Scales numerical features using `StandardScaler` to optimize model training.
- Saves the fitted scaler to `models/scaler.pkl`.

4. Model Training & Evaluation

The pipeline trains and benchmarks five separate classifiers:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)

Models are evaluated on Accuracy, Precision, Recall, and F1-Score. The code automatically isolates the best-performing model based on testing accuracy.

5. Optimization & Final Metrics

The top-performing model undergoes hyperparameter tuning using `GridSearchCV` with 5-fold cross-validation. Performance diagnostics include:

- **Classification Report:** Detailed breakdown of precision, recall, and f1-score.
- **Confusion Matrix:** Map of true vs. predicted classifications.
- **ROC-AUC Curve:** Calculation of the Area Under the Curve to measure classification thresholds.

The final optimized model is saved as `models/heart_disease_model.pkl`.

Setup & Requirements

Installation

Clone the repository and install the required libraries:

```
git clone https://github.com/your-username/Heart-Disease-Prediction.git
cd Heart-Disease-Prediction
pip install -r requirements.txt
```

Dependencies

- pandas

- numpy
- matplotlib
- scikit-learn
- joblib